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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Branch	CSE
Course Outcome	CO6 (0/1 Knapsack using Dynamic Programming)

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using 0/1 Knapsack

Work Done Summary:

The Waste Sense system uses the 0/1 Knapsack algorithm to optimize waste collection vehicle loading. Each waste bin contains a certain amount of recyclable material and has an associated value representing its recycling importance. Since collection vehicles have limited capacity, the system selects the most valuable combination of waste bins while staying within the vehicle's capacity constraints.

Implementation Details:

1. Create a list of waste bins with weights and values.
2. Define the vehicle capacity.
3. Apply Dynamic Programming (Memorization or Tabulation).
4. Calculate the maximum recyclable value obtainable.
5. Identify the optimal set of bins for collection.
6. Display the selected bins and total value.

Result: Screenshot/output:

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Maximum Total Value = 15
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Selected Bins:
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Bin 4 (Weight = 5 kg, Value = 8)
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Bin 2 (Weight = 3 kg, Value = 4)
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Bin 1 (Weight = 2 kg, Value = 3)
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Example: Waste Bins

Bin (Item)	Weight (kg)	Value (Importance)
Bin 1	2	3
Bin 2	3	4
Bin 3	4	5
Bin 4	5	8
Bin 5	9	10

Vehicle Capacity (W) = 10 kg

Goal:
Select the combination of bins with total weight ≤ 10 kg that gives the maximum total value.

0/1 Knapsack using Dynamic Programming

Let:

- n = number of bins (items) = 5
- W = vehicle capacity = 10
- wt[i] = weight of ith bin
- val[i] = value of ith bin

DP State:
dp[i][w] = maximum value achievable using first i items (without repetition) and capacity w.

DP Relation:

$$dp[i][w] = \max(dp[i-1][w], \text{val}[i-1] + dp[i-1][w - \text{wt}[i-1]])$$
 if $\text{wt}[i-1] \leq w$
 else $dp[i][w] = dp[i-1][w]$

Base Case:
 $dp[0][w] = 0$ for all w
 $dp[i][0] = 0$ for all i

DP Table Construction (Rows: Items 0...5, Columns: Capacity 0...10)

i \ w	0	1	2	3	4	5	6	7	8	9	10
0 (No item)	0	0	0	0	0	0	0	0	0	0	0
1 (Bin1: w=2, v=3)	0	0	3	3	3	3	3	3	3	3	3
2 (Bin2: w=3, v=4)	0	0	3	4	4	7	7	7	7	7	7
3 (Bin3: w=4, v=5)	0	0	3	4	5	7	8	9	9	12	12
4 (Bin4: w=5, v=8)	0	0	3	4	5	8	8	11	12	12	13
5 (Bin5: w=9, v=10)	0	0	3	4	5	8	8	11	12	12	13

Legend:

- Rows show items considered (0 to 5).
- Columns show capacity from 0 to 10 kg.
- dp[5][10] = 13 → Maximum value for capacity 10 kg.

Backtracking to Find Selected Bins

Start from dp[5][10] and move upwards:

- dp[5][10] = 13 = dp[4][10] → Bin5 not taken (9 kg).
- dp[4][10] = 13 ≠ dp[3][10] = 12 → Bin4 taken (5 kg).
Remaining capacity = 10 - 5 = 5
- dp[3][5] = 7 ≠ dp[2][5] = 4 → Bin3 taken (4 kg).
Remaining capacity = 5 - 4 = 1
- dp[2][1] = 0 = dp[1][1] = 0 → Bin2 not taken.
- dp[1][1] = 0 = dp[0][1] = 0 → Bin1 not taken.

Selected Bins: {Bin 3, Bin 4}

Total Weight = 4 + 5 = 9 kg (≤ 10 kg)
Total Value = 5 + 8 = 13 (Maximum)

All Possible Combinations (For Reference)

Combination	Total Weight (kg)	Total Value	Valid (≤10 kg)?
{}	0	0	Yes
{Bin1}	2	3	Yes
{Bin2}	3	4	Yes
{Bin3}	4	5	Yes
{Bin4}	5	8	Yes
{Bin5}	9	10	Yes
{Bin1, Bin2}	5	7	Yes
{Bin1, Bin3}	6	8	Yes
{Bin1, Bin4}	7	11	Yes
{Bin2, Bin3}	7	9	Yes
{Bin2, Bin4}	8	12	Yes
{Bin3, Bin4}	9	13	Yes ← Best
Other combinations	>10	-	No

Result: Optimal Selection

Select bins that give maximum recycling importance without exceeding the vehicle capacity.

Optimal Bins to Collect

Bin 3 → Weight = 4 kg, Value = 5
Bin 4 → Weight = 5 kg, Value = 8

Total Weight = 9 kg (≤ 10 kg)
Maximum Total Value = 13

- The Waste Sense system successfully applies the 0/1 Knapsack algorithm using Dynamic Programming to select the most valuable combination of recyclable waste bins within vehicle capacity limits. The solution improves collection efficiency and maximizes recycling output.

GIT HUB LINK : <https://github.com/Mahaiswaryakaruturi/DSA2-Case-Studies>

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